



EXTREMAL GRAPH THEORY

Erdős Problem 915

From Proof to Search

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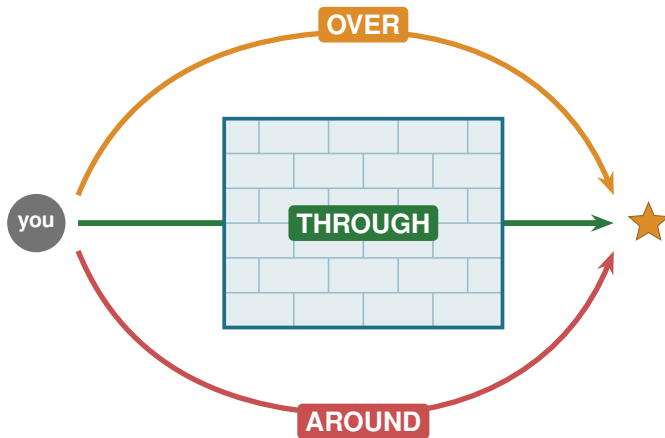
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KU LEUVEN

A hard problem is a wall



Three ways past the wall

THROUGH

Proof

one argument,
every n at once

OVER

Calculation

a finite check,
with a certificate

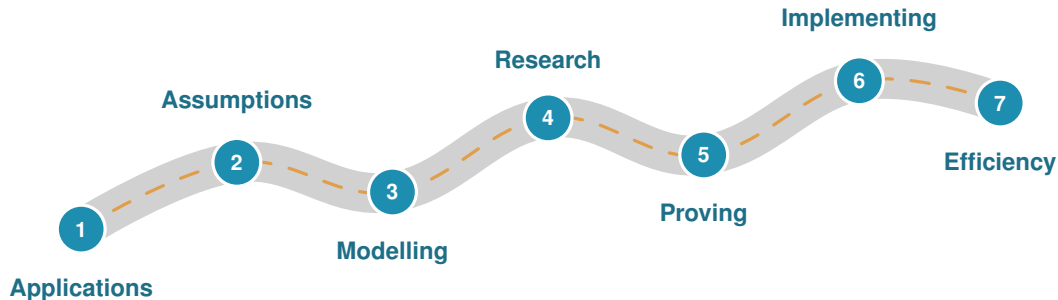
AROUND

Approximation

search for examples,
a lower bound

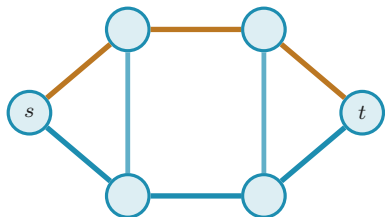
This thesis uses all three.

The road of a research problem



The slides ahead follow this road.

Problem 915, precisely



two independent s - t routes

A graph: n vertices, E edges.

Routes. $\lambda(u, v)$ = independent u - v paths.

Rule. No pair gets m routes: $\lambda^{\max} \leq m - 1$.

Goal. Pack in as many edges as possible.

$$\ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor \quad (\text{Mader})$$

What is it good for?

No direct application. The methods are the point.

The question

Quantify a network's
spare, independent
routes.

The tools

maximum flow,
search, certificates

Where it lives

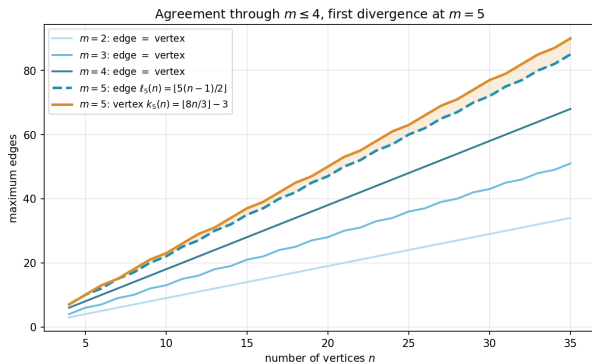
grids, pipelines,
road networks

One question, twelve variants

Model	simple	multigraph	hypergraph	$\times 3$	
Direction	undirected	directed		$\times 2$	= 12
Separation	edge-disjoint	vertex-disjoint		$\times 2$	

Mader's theorem is one corner. The other eleven are the journey.

Edge routes versus vertex routes



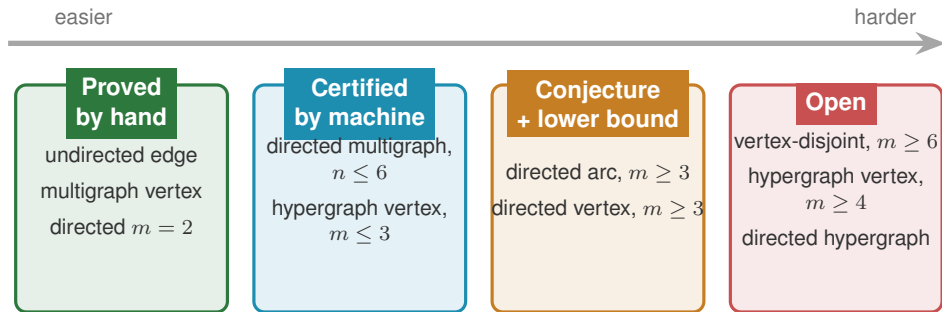
λ : edge-disjoint. κ : vertex-disjoint.

$m \leq 4$ same answer (Leonard).

$m = 5$ first gap: $k_5(n) = \lfloor \frac{8n}{3} \rfloor - 3$.

$m \geq 6$ open since 1974.

What fell, and how hard



How the machine works

Search: annealing vs tabu

variant	opt.	SA	tabu
simple undirected $n=7$	9	9	9
simple directed $n=7$	18	16	18
directed multi $n=7$	24	20	24

Tabu reaches the directed optima where annealing plateaus.

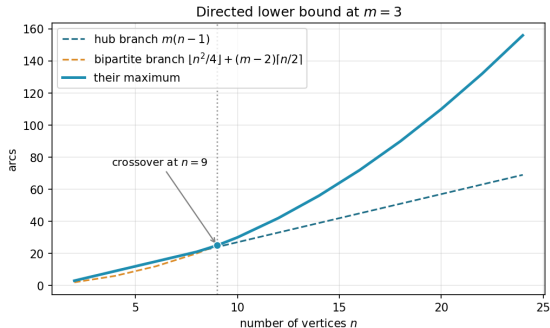
Code: Python with a C hot path

Python core: `numpy + scipy max-flow`

optional C hot path (`_erdos_fast.so`)

Speedups: `max-flow` $\approx 2\times$, canonical form **147** $\times$.

Results on the directed frontier



At $m = 3$: linear hub leads early,
quadratic family overtakes near $n = 9$.

Two genuine advances

solved Hypergraph vertex, $m \leq 3$:

$$k_m^{(r)}(n) = \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor.$$

reduced Directed multigraph,

$m = 3$:

the whole family collapses to **two checks** on 7 vertices.

The whole landscape

Separation	Model	Value	Status
Undirected			
edge	simple	$\lfloor m(n-1)/2 \rfloor$	proved (Mader)
edge	multi	$(m-1)(n-1)$	proved
edge	hyper	$\lfloor (m-1)(n-1)/(r-1) \rfloor$	proved (Gomory-Hu)
vertex	simple	$\lfloor m(n-1)/2 \rfloor$ ($m \leq 4$), $\lfloor 8n/3 \rfloor - 3$ ($m=5$)	cited, open $m \geq 6$
vertex	multi	parallel edges do not help	proved
vertex	hyper	$\lfloor (m-1)(n-1)/(r-1) \rfloor$ ($m \leq 3$)	proved $m \leq 3$, open $m \geq 4$
Directed			
arc	simple	$\max(2(n-1), \lfloor n^2/4 \rfloor)$ ($m=2$)	proved, conj. $m \geq 3$
arc	multi	$(m-1) \max(2(n-1), \lfloor n^2/4 \rfloor)$	proved $n \leq 6$, cond. $m=3$
arc	hyper	quadratic in n , exact value open	lower bound
vertex	simple	$\max(2(n-1), \lfloor n^2/4 \rfloor)$ ($m=2$)	proved, conj. $m \geq 3$
vertex	multi	reduces to simple digraph	proved
vertex	hyper	quadratic in n , exact value open	lower bound

proved

certified

conjecture

open

What is left, and why it matters

Open problems

- the backward-arc lemma (closes the directed case)
- two finite $n = 7$ checks ($m = 3$ multigraph)
- vertex-disjoint $m \geq 6$, open since 1974
- hypergraph vertex $m \geq 4$

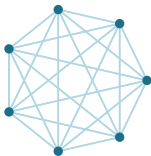
One pipeline, every variant

- one representation
- one exact checker
- one certifier
- one search

The microscope does not answer every question. It shows exactly which questions are left.

Thank you

Questions?

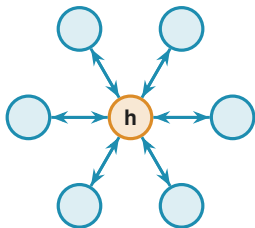


Backup slides follow.

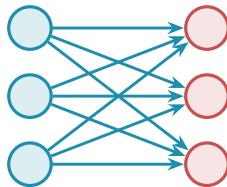
BACKUP SLIDES

Dive deeper into any single variant.

Backup: the directed arc conjecture



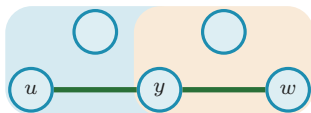
bidirected hub: $2(n - 1)$



bipartite digraph: $\lfloor n^2/4 \rfloor$

At $m = 2$ the value $\max(2(n - 1), \lfloor n^2/4 \rfloor)$ is proved (the two constructions tie at $n = 7$). For $m \geq 3$ the conjecture adds the augmented-bipartite family. The missing piece is the **backward-arc lemma** for the matching upper bound.

Backup: the hypergraph vertex problem



a Berge path u, e_1, y, e_2, w

Routes are **Berge paths**, alternating vertices and hyperedges.

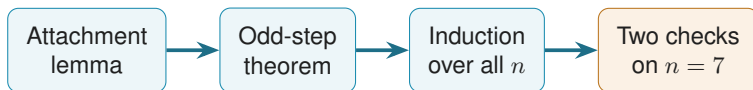
Solved ($m \leq 3$, every r , all n):

$$k_m^{(r)}(n) = \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor,$$

via a rank bound on incidence graphs (Tutte SPQR, valid where $\kappa \leq 2$).

Open at $m \geq 4$: needs a 4-connectivity analogue.

Backup: collapsing an infinite family to two checks

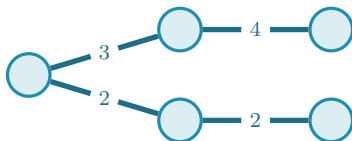


The two remaining facts (multiplicities in $\{0, 1, 2\}$):

- (a) $L_3^{\text{dir}}(7) = 24$;
- (b) the only 24-arc extremisers are $2B_{3,4}$, $2B_{4,3}$, and the doubled path P_7 .

The wall here is runtime, not structure.

Backup: how connectivity is stored



A **Gomory-Hu tree** stores all $\binom{n}{2}$ connectivities in $n - 1$ numbers: $\lambda(u, v)$ is the lightest edge on the tree path, and $\lambda^{\max}$ is the heaviest tree edge.

Backup: values at a glance

Object	Extremal value
simple undirected edge, $\ell_m(n)$	$\lfloor m(n-1)/2 \rfloor$
multigraph edge, $L_m(n)$	$(m-1)(n-1)$
hypergraph edge, $\ell_m^{(r)}(n)$	$\lfloor (m-1)(n-1)/(r-1) \rfloor$
simple vertex, $k_m(n)$ ($m \leq 4$)	$\lfloor m(n-1)/2 \rfloor$
simple vertex, $k_5(n)$	$\lfloor 8n/3 \rfloor - 3$
directed arc, $m = 2$	$\max(2(n-1), \lfloor n^2/4 \rfloor)$
hypergraph vertex, $m \leq 3$	$\lfloor (m-1)(n-1)/(r-1) \rfloor$
random-graph threshold	$p^* = m/n$